Unit 5: Quantitative Research Design and Methodology

Lesson 4

Planning Data Collection and Analysis

Learning Objectives

At the end of the lesson, you should be able to do the following:

- Identify the different data collection methods for quantitative research.
- Evaluate the selected data collection method in a quantitative research.
- Apply appropriate data analysis based on the data collection method used.

Brilliant Minds Preschool Student Checklist

___ Can identify the colors properly

___ Can follow simple instructions

___ Can write their own name

___ Identifies all the phonological sounds of the alphabet

___ Can match uppercase and lowercase letters

___ Can recite the alphabet song without difficulty

Data Collection Methods

- Questionnaires
- Experiment
- Quantitative Observation
- Quantitative Interview

What are some differences between data collection procedures in quantitative and qualitative research?

Questionnaires

A questionnaire is composed of a **written list of item questions** wherein the participants can answer independently.

Experiment

- An experiment is designed to **test hypotheses** under controlled conditions and involves the manipulation of variables.
- Three types of experiments: **laboratory**, **field**, and **natural.**

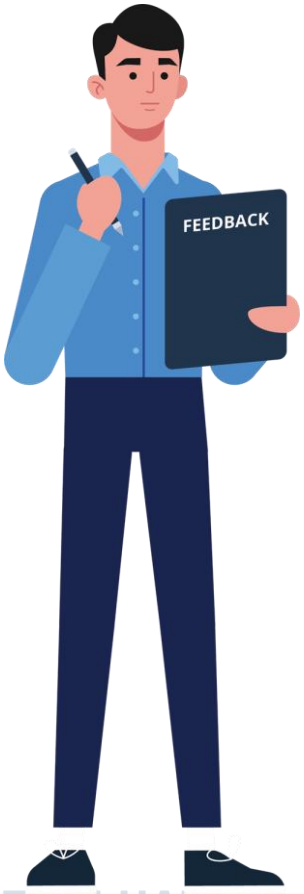
Quantitative Observation

- Observation is a process wherein researchers purposely or selectively watch or listen to an o**ccurring interaction or phenomenon**.
- Quantified by using **rating scales** and **checklists**.

Quantitative Interview

Quantitative interviews are also called standardized interviews or survey interviews because of their similarity to a survey's **question-and-answer** format.

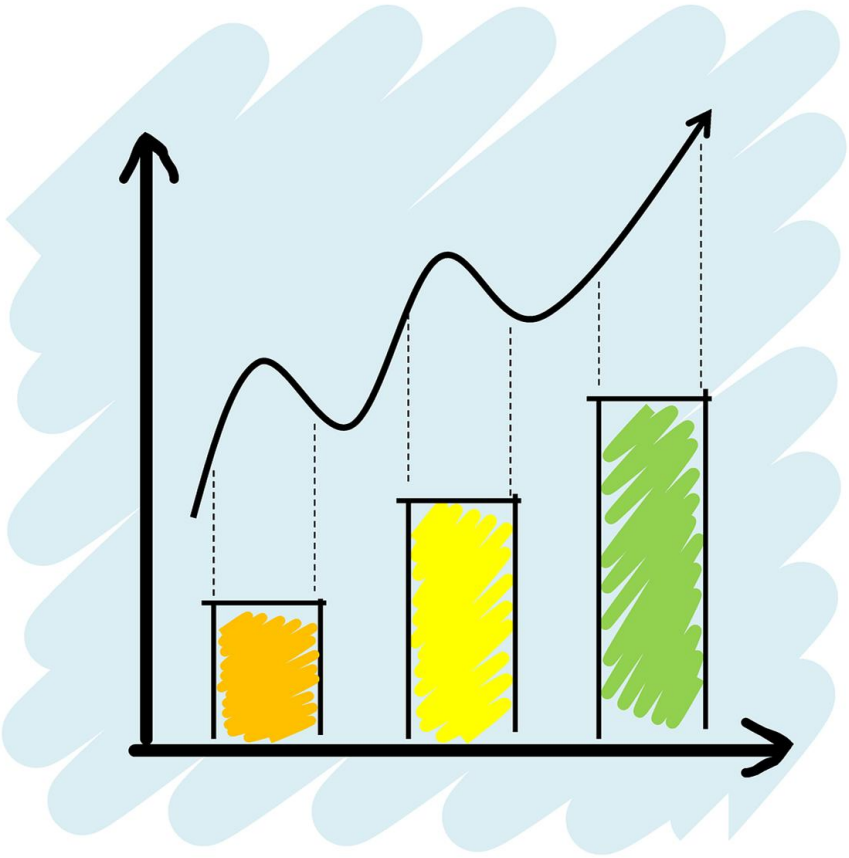
Evaluating Data Collection Methods



- Data collection procedure must be **ethical.**
- Data collection procedures must be **standardized.**
- Identify how the researchers **reduced threats.**

Why do research studies need to be ethical?

Introduction to Data Analysis and Hypothesis Testing



Quantitative data analysis is a powerful tool that is often associated with large-scale research.

Scales of Data

Scale of Data	Definition	Example
Nominal	A nominal scale pertains to categories and has no numerical meaning.	In a demographic form, “1” might denote female and “2” denote male.
Ordinal	In ordinal scales, the data are classified according to the ranking or order of the variables.	A software company may ask its clients: How would you rate our app? ● Excellent ● Very Good ● Good ● Bad ● Poor

Scales of Data

Scale of Data	Definition	Example
Interval	An interval scale assumes that there is an equal interval between each data point.	Measuring temperature using the Fahrenheit scale.
Ratio	A ratio scale indicates that there is an absolute zero. The characteristics of the three scales above are also found in the interval scale.	Measures of distance, price of goods, and weight.

Descriptive and Inferential Statistics

Descriptive

simply **summarizes** the data by presenting it into more manageable bits

Inferential

enables researchers to **test their hypotheses** and **conduct regression analysis**

Tip

Numerical data from quantitative research can be analyzed or performed using computer programs.

Hypothesis Testing

- **Hypothesis testing** allows researchers to test their assumptions.
- The two types of hypotheses are the **null hypothesis** and the **alternate hypothesis**.

Activity

Step 1: Data Collection Procedure
Step 2: Indicators of a Good Data Collection Procedure
Step 3: Data Analysis

Identify the Data Collection Method

Lyn is a developmental psychologist who wants to study the effects of early exposure to gadgets of children ages one to three years old. She developed a tool to note remarkable behaviors of her participants.

Identify the Data Collection Method

Athena uses e-commerce to sell beauty products. She wanted to know how satisfied her customers are when it comes to the service they provide and the quality of their products. She prefers to use an online tool where her customers can respond on their own.

Quick Check

How important is the evaluation of the data collection procedure of a study?

The most common types of data collection methods used in quantitative research are **questionnaires**, **experiments**, **quantitative observations**, and **quantitative interviews**.

Wrap Up

- Data collection methods must be evaluated by its **ethical considerations**, **standardization**, and how the researchers **reduced threats** to their data.

- The scales of data arranged according to complexity are **nominal**, **ordinal**, **interval**, and **ratio scale**.

Numerical data in quantitative research can be summarized or presented using either **descriptive** or **inferential statistics.**

Wrap Up

Quantitative Research

Data Collection Procedure

- Questionnaires
- Experiments
- Quantitative Observation
- Quantitative Interview

Evaluation

- Ethical considerations
- Standardization
- Reduction of threats

Data Analysis

- Scales of Data
- Descriptive Statistics
- Inferential Statistics
- Hypothesis Testing

Bibliography

"Analyzing Quantitative Research." Center for Innovation in Research and Teaching. Grand Canyon University. Accessed June 17, 2020. https://cirt.gcu.edu/research/developmentresources/research_ready/quantresearch/analyze_data.

Belyh, Anastasia. "Overview of Qualitative And Quantitative Data Collection Methods." Cleverism.com. February 26, 2017. https://www.cleverism.com/qualitative-and-quantitative-data-collection-methods/.

Blackstone, Amy. *Principles of Sociological Inquiry: Qualitative and Quantitative Methods.* Washington: Saylor Foundation, 2012.

Cohen, Louis, Manion, Lawrence, and Keith Morrison. *Research Methods in Education*. 8th ed. New York: Routledge, 2018.

Kumar, Ranjit. *Research Methodology: A Step-by-step Guide for Beginners.* 3rd ed. London: SAGE Publications, Ltd., 2011.

Langdridge, Darren, and Gareth Hagger-Johnson. *Introduction to Research Methods and Data Analysis in Psychology.* 3rd ed. England: Pearson, 2013.

Leary, Mark R. *Introduction to Behavioral Research Methods.* 6th ed. New Jersey: Pearson, 2012.

Plano Clark, Vicki L., and John W. Creswell. *Understanding Research: A Consumer's Guide.* 2nd ed. New Jersey: Pearson, 2015.